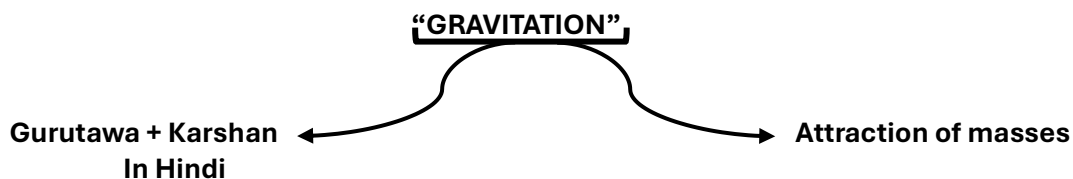


GRAVITATION

- **Introduction:**



- **Gravitation / Gravitational Forces**

- "Every object / body in the universe attracts every other object with a certain force. This certain force is known as Gravitational Force."

For example:-

- ✓ An apple / any fruit falls down from a tree by a gravitational force on earth.
- ✓ Rain falls down from the sky to the earth due to the gravitational force on earth.
- ✓ Water from mountain flow down the rivers.
- ✓ The earth / All planets move around the sun due to gravitational force of Sun.
- ✓ The moon moves around the earth due to gravitational force of earth.

- **Gravity of earth / Gravitational force of earth**

- The earth attracts all objects towards its centre with a certain force. This force is called Gravity of earth / Gravitational force of earth.

OR

In the gravitational force, If earth is one of the two objects then the gravitational force is known as gravitational force of earth / gravity of earth.

Note:

- "The gravitational force" is also called "gravity" is always a force of attraction between two objects.

For example:-

- ✓ When an object is thrown vertically upwards it falls towards the earth.
- ✓ When an object released from some height it falls towards the centre.

Note:

- The gravity of earth is a pull force towards the centre of the earth.

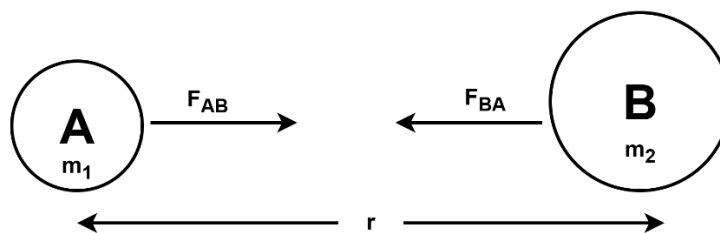
- **The universal law of gravitation**

- The law proposed by Sir Issac Newton due to this, this law is also called Newton's Fourth law OR Newton's law of gravitation. The universal law of gravitation is defined as, "Every body (object) in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to square of distance between them".

Note:

- The direction of the force is along the line joining the two objects (bodies).

Let A and B are two objects of masses m_1 and m_2 kg are separated by distance 'r' to each other.
Let the attraction forces between the objects A and B is F.



▪ **According to the universal law of gravitation**

a) The force between two bodies is directly proportional to the product of their masses.

i.e, $F \propto m_1 m_2$ (i)

b) The force between two bodies is inversely proportional to the square of the distance between them.

i.e, $F \propto \frac{1}{r^2}$ (ii)

On combining equation (i) & (ii) both sides.

$$F \propto \frac{m_1 \cdot m_2}{r^2}$$

$$F = G \cdot \frac{m_1 \cdot m_2}{r^2}$$

Where, G is a constant which is known as “Universal gravitational constant”.

OR

$$G = \frac{F \times r^2}{m_1 \times m_2}$$

Note:

➤ r is the distance between centres of both bodies.

→ The S.I unit of Gravitational constant is Nm^2/kg^2 .

$$\begin{aligned} \therefore \text{S.I unit of } G &= \frac{\text{S.I unit of } F \times \text{S.I unit of } r^2}{\text{S.I unit of } m_1 \text{ and } m_2} & \left(\because G = \frac{F \times r^2}{m_1 \times m_2} \right) \\ &= \frac{\text{N} \times \text{m}^2}{\text{kg} \times \text{kg}} &= \text{Nm}^2/\text{kg}^2 \end{aligned}$$

→ The value of $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

↓
This value was experimentally measured in the laboratory by Cavandish, long after Newton's death.

↓
Full name – Henry Canvadish an English Philosopher and Scientist.

Note:

1. “G” is a universal constant its value same at every planet, every place (any where) like as space etc.
 G does not depend on the mass of the body, distances between bodies. i.e, it is fixed for every object / body.
 So, $G = 6.67 \times 10^{-11} \text{ Nm}^2 / \text{Kg}^2$ is fixed.

2. If two bodies of masses $m_1 = 1 \text{ kg}$, $m_2 = 1 \text{ kg}$ distance between them is 1 m then the gravitational force between them is

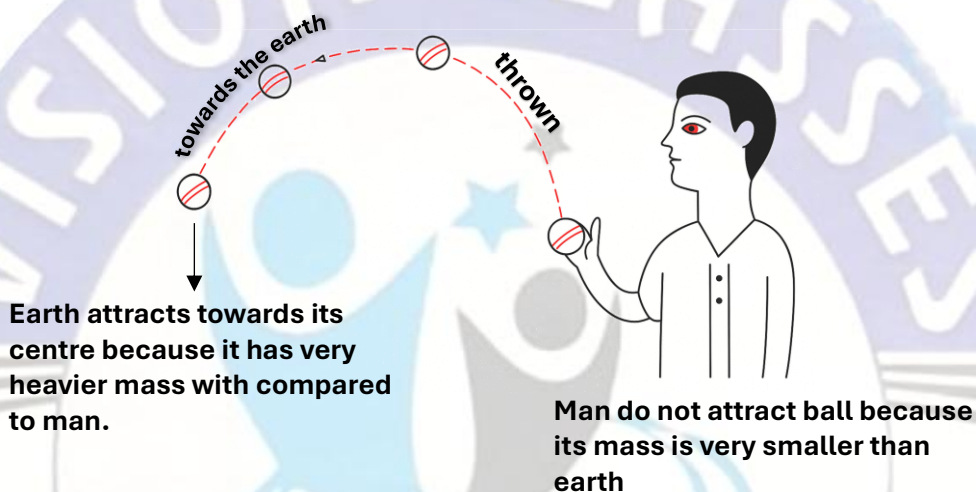
$$\therefore F = G \times \frac{m_1 \times m_2}{r^2}$$

$$\therefore F = \frac{6.67 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2} \times 1 \text{ kg} \times 1 \text{ kg}}{1 \text{ m} \times 1 \text{ m}}$$

$\Rightarrow F = 6.67 \times 10^{-11} \text{ N}$, is equal to the gravitational constant.

Very – Very small force

3. The several objects on the earth attract to each – other constantly, but they do not motion because the gravitational force is very small.



Special Note:

a) $\therefore F \propto m_1 \times m_2$

i.e, $F \propto m_1$ & $F \propto m_2$

Conclusion (a) If one of mass of object increases then F increases.

- b) If both of masses of objects increases, then F increases in large amount.

- c) If one of mass of object is very – very large (like – earth, sun etc.) then other object of lighter mass moves towards the heavier object.

For example:-

✓ Apple falls down from tree towards the earth.

Q. Calculate the force of gravitation due to earth on a ball of mass 1 kg lying on the earth.

Ans $\therefore m_1 = 1 \text{ kg}$

$m_2 = 6 \times 10^{24} \text{ Kg}$

$R = 6.4 \times 10^6 \text{ m}$

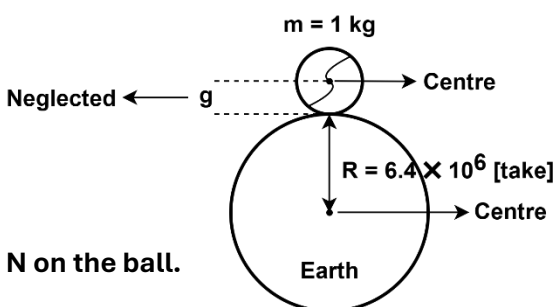
$G = 6.67 \times 10^{-11} \text{ Nm}^2 / \text{kg}^2$

$\therefore F = G \times \frac{m_1 \cdot m_2}{R^2}$

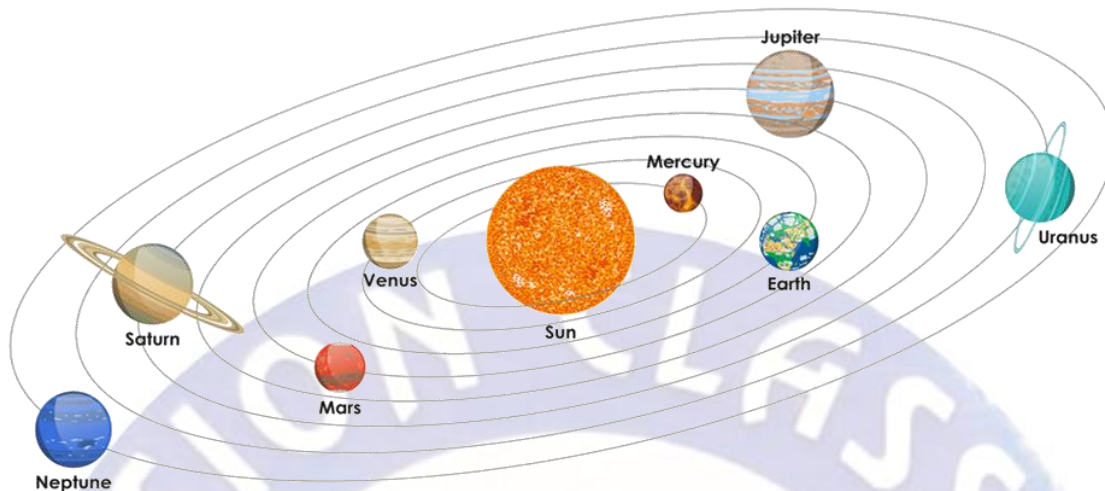
$\Rightarrow F = \frac{6.67 \times 10^{-11} \times 1 \times 6 \times 10^{24}}{6.4 \times 6.4 \times 10^6 \times 10^6} = 9.8 \text{ N}$

$\Rightarrow F = 9.8 \text{ N}$

$\therefore$ Thus, the earth exerts a gravitational force of 9.8 N on the ball.



- **All planets move around the sun due to gravitational force, but sun do not move any one because the sun has the heaviest mass.**



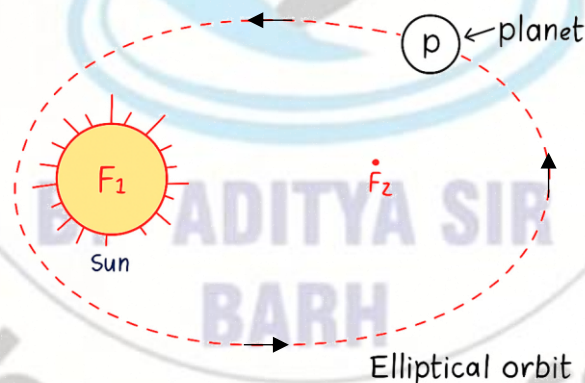
- **Kepler's law of Planetary motion**

- In 16th century, Johannes Kepler an astronomer who established three laws which is related to motion of the planets, these laws are known as Kepler's laws of Planetary motion.
- Kepler's law of planetary motion is also applicable for satellites like as moon.

- **There are three laws of Kepler of Planetary motion**

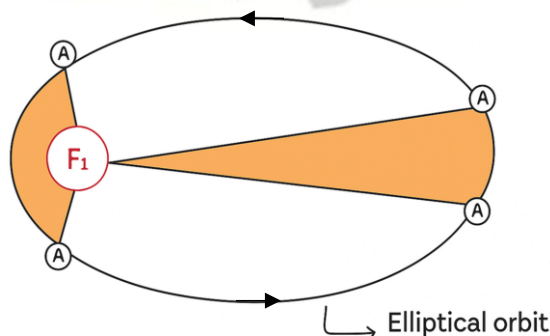
1. **Kepler's first law of Planetary motion**

- All the planets move in elliptical orbits around the sun, with the sun at one of the two foci of the elliptical orbit.



2. **Kepler's second law of Planetary motion**

- Each planet revolves around the sun in such a way that the line joining the planet to the sun sweeps over equal areas in equal time intervals of time.



i.e, A planet moves faster when it is closer to the sun and moves slowly when it is farther from the sun. so, the speed of planet is not constant.

3. Kepler's third law of planetary motion

- The cube of the mean distance of a planet from the sun is directly proportional to the square of time it takes to move around the sun.

i.e, $r^3 \propto T^2$
 $\rightarrow$ D.P

$$\Rightarrow r^3 = \text{Constant} \times T^2$$

$$\Rightarrow \frac{r^3}{T^2} = \text{Constant}$$

Where,

r = mean distance of a planet from the sun.

& T = time period of a planet around the sun.

Note:

- All planets move around the sun due to gravitation force which the sun exerts on them.

• Inverse square law / rule

- This law is actually derived by Newton's Universal law of gravitation, According to this law, "The force between two bodies is inversely proportional to the square of distance between them is called the inverse – square rule."

i.e, $F \propto \frac{1}{r^2}$

Newton derived the inverse square rule for gravitational force.

Let a planet of mass m kg, moving with speed (velocity) v around a circular orbit of radius r .

A centripetal force F acts on the orbiting planet.

$$\therefore F = \frac{mv^2}{r}$$

$$\Rightarrow F \propto \frac{v^2}{r} \quad \dots\dots\dots (i) \quad [\because \text{mass of planets is constant}]$$

$$\therefore V = \frac{2\pi r}{T} \quad [\text{For circular path}]$$

$$\Rightarrow V \propto \frac{r}{T} \quad [\because 2\pi \text{ is constant}]$$

On squaring both sides,

$$V^2 \propto \frac{r^2}{T^2}$$

$$\Rightarrow V^2 \propto \frac{r^3}{T^2} \times \frac{1}{r} \quad [\text{On multiplying by } r \text{ in numerator and denominator}]$$

$$\Rightarrow V^2 \propto \frac{1}{r} \quad [\because \frac{r^3}{T^2} \text{ is constant}]$$

Putting $\frac{1}{r}$ in place of V^2 in equation (i)

$$\Rightarrow F \propto \frac{1}{r \times r}$$

$$\Rightarrow F \propto \frac{1}{r^2}$$

Proved.

- **Free fall**

- A falling body with initial velocity zero towards the earth is called free fall and the body is known as free fall body.
- **According to Galileo Galilei (1564 – 1642) [A famous Italian Scientist]**
- The acceleration of the free fall objects or, the falling objects towards the earth is free from the mass.

For example:-

- ✓ If two objects of masses 60 kg and 100 kg drops from top of a tower then both objects heat the ground at same time.

Exception:

- In the case of feather and hammer. Hammer heats the ground first because its cross section area less with respect to the feather.

⇒ Robert Boyle proved that any objects having any cross section area both heat the ground at same time in vacuum.

His experiment also proved that the acceleration of the falling bodies independent from their masses.

- **Vacuum**

- That place where air (oxygen) is not present, called vacuum.
OR
An empty space in which there is no air as well as other gases is called vacuum.

For example:-

- ✓ Space.
- **Acceleration due to Gravity**
- The acceleration created in a falling body due to attraction of earth towards the centre is called acceleration due to gravity.
- The acceleration due to gravity is denoted by 'g'.
- The value is 9.8 m/s^2 .
- The 'g' of free-falling body is uniform.
- When an object in free fall then it is accelerated with 9.8 m/s^2 (+ve).
i.e, $g = +ve$
- When an object thrown vertically upward it retards with 9.8 m/s^2 .
i.e, $g = -ve$

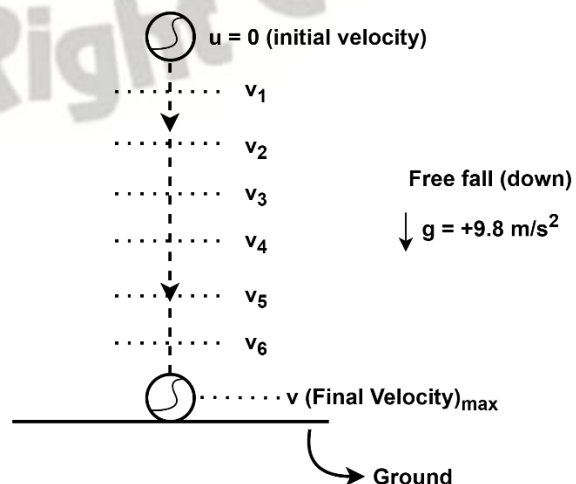
Case 1:

Note:

- **Velocity:**

$$u < v_1 < v_2 < v_3 < v_4 < v_5 < v_6 < v$$

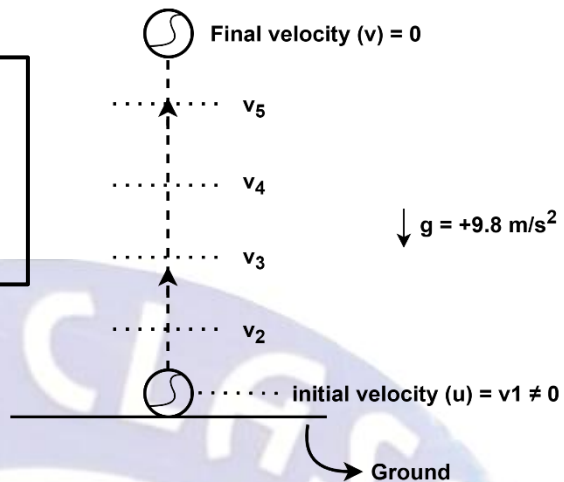
↘ maximum



Case 2: When object thrown upward vertically

Note:

➤ **Velocity:**
 $v_1 > v_2 > v_3 > v_4 > v_5 > v$
 minimum

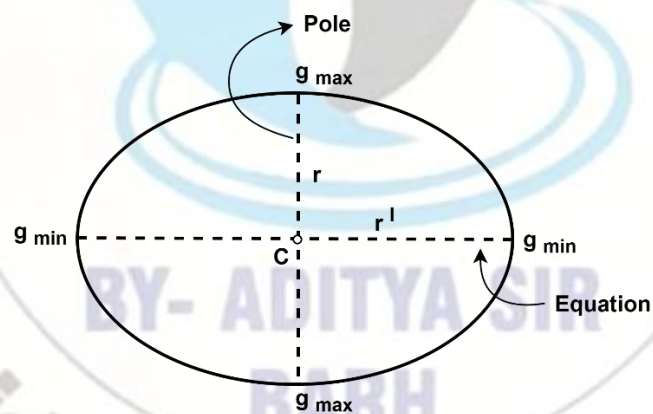


Special Note:

- Free fall of an object is a type of uniformly accelerated motion.

Note:

- $g = 9.8 \text{ m/s}^2$
 $= 10 \text{ m/s}^2$



EARTH

Note:

1. The value of acceleration due to gravity 'g' at the pole of the earth is maximum because the distance from centre to the surface is minimum.
2. The value of g is minimum at equator because the distance from centre to surface is maximum.
3. The value of 'g' is zero at centre because distance is zero.
4. The value of 'g' variable at the different places of the earth thus, the value of g is also different at every planets.

For example:-

- ✓ The value of g on earth = 9.8 m/s^2
 The value of g on moon = 1.63 m/s^2

• What are the differences between g & G

1. G is a universal constant while 'g' is acceleration due to gravity.
2. The value of 'G' is constant at everywhere while the value of 'g' is variable at different places.
3. The value of 'G' can't be zero while the value of 'g' may be zero.
4. The S.I unit of 'G' is Nm^2/kg^2 while S.I unit of 'g' is m/s^2 .
5. 'G' is a scalar quantity while 'g' is a vector quantity.
6. The value of 'G' = $6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ while the value of $g = 9.8 \text{ m/s}^2$ at the earth surface.

• The relation between 'G' and 'g'

We know that $F = ma$

$$\Rightarrow F = mg \dots\dots\dots (i) \quad [\because a = g]$$

&

$$F = G \cdot \frac{M \times m}{R^2} \dots\dots\dots (ii)$$

where,

m = mass of object
 M = mass of earth
 R = radius of the earth

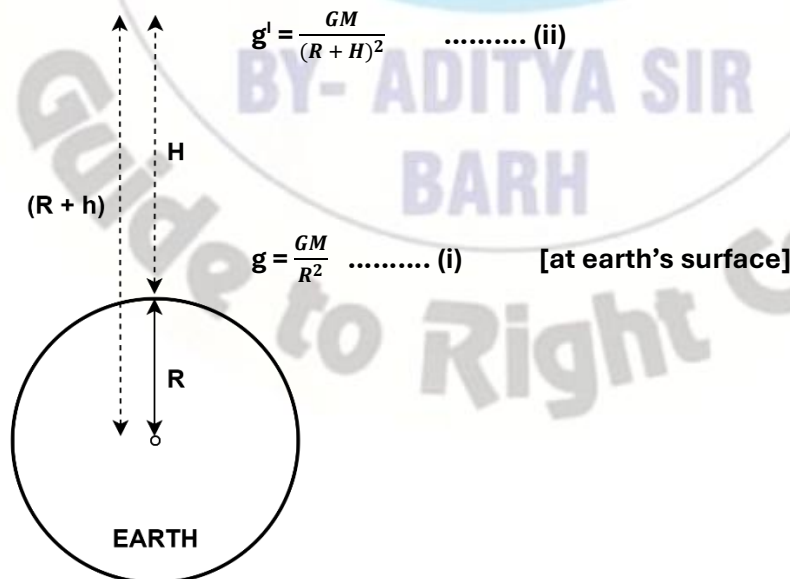
From equation (i) & equation (ii)

$$mg = G \cdot \frac{m \times M}{R^2}$$

$$\Rightarrow \boxed{g = \frac{GM}{R^2}}$$

which is required relation between g and G

• The value of 'g' at H height above the earth's surface



On dividing equation (ii) by equation (i)

$$\frac{g'}{g} = \frac{\frac{GM}{(R + H)^2}}{\frac{GM}{R^2}}$$

$$\Rightarrow \frac{g'}{g} = \frac{R^2}{(R+H)^2}$$

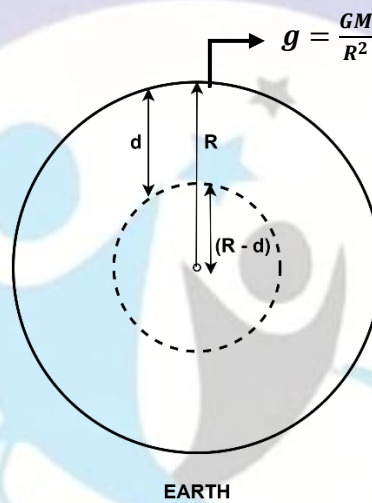
$$\Rightarrow g' = g \times \left\{ \frac{R}{R+H} \right\}^2 = g \times \frac{1}{\left(\frac{R+H}{R} \right)^2}$$

$$\Rightarrow g' = g \times \left(\frac{1}{\left(\frac{R}{R} + \frac{H}{R} \right)^2} \right)$$

$$\Rightarrow g' = g \times \left(\frac{1}{\left(1 + \frac{H}{R} \right)^2} \right)$$

$$\Rightarrow \boxed{g' = g \times \left(1 + \frac{H}{R} \right)^{-2}}$$

- The value of 'g' under the earth at 'd' depth



$$\therefore g \text{ at the surface of earth} = \frac{GM}{R^2}$$

Let density of earth is ρ .

$$\therefore g = \frac{G}{R^2} \times M \quad [\text{On earth's surface}]$$

$$\Rightarrow g = \frac{G}{R^2} \times \frac{4}{3} \pi R^3 \times \rho \quad [M = V \times \rho]$$

$$\Rightarrow g = \frac{4}{3} \pi \times GR \times \rho \quad \dots\dots\dots (i)$$

and the value of g at 'd' depth

$$g' = \frac{G}{(R-d)^2} \times \frac{4}{3} \pi (R-d)^3 \times \rho$$

$$\Rightarrow g' = G(R-d) \times \frac{4}{3} \pi \times \rho \quad \dots\dots\dots (ii)$$

On dividing equation (ii) by equation (i)

$$\Rightarrow \frac{g'}{g} = \frac{\frac{4}{3} \pi \times G (R-d) \times \rho}{\frac{4}{3} \pi \times GR \times \rho}$$

$$\Rightarrow \frac{g'}{g} = \frac{R-d}{R} = \frac{R}{R} - \frac{d}{R}$$

$$\Rightarrow \boxed{g' = g \left(1 - \frac{d}{R}\right)}$$

→ which is required value of 'g' under 'd' depth under the earth surface.

Q. At what distance above the earth surface the value of g is $\frac{g}{2}$.

Ans ∴ $g' = \frac{g}{2}$

& $g' = g \left(1 + \frac{H}{R}\right)^{-2}$

$\Rightarrow \frac{g}{2} = g \left(1 + \frac{H}{6400}\right)^{-2}$ [$\because R = 6400$]

$\Rightarrow \frac{1}{2} = \frac{1}{\left(1 + \frac{H}{6400}\right)^2}$

$\Rightarrow \left(1 + \frac{H}{6400}\right)^2 = 2$

$\Rightarrow 1 + \frac{H}{6400} = \sqrt{2} \Rightarrow 6400 + H = 6400\sqrt{2}$

$\Rightarrow H = 6400\sqrt{2} - 6400$

$\Rightarrow H = 6400(\sqrt{2} - 1)$

$\Rightarrow H = 6400(1.414 - 1)$

$= 6400 \times 0.414$

$\Rightarrow H = 2650 \text{ Km}$ (Approx)

The required height of 2650 km.

▪ **The difference between mass and weight**

Mass	Weight
Mass is amount of substance of an object. i.e, $m = \frac{\text{Force}}{\text{Acceleration}}$	Weight is the pull force exerted by earth on an object towards its centre. i.e, $w = m \times g$
It is a scalar quantity because it has only magnitude.	It is a vector quantity because it has magnitude as well as direction.
It is measured by balance scale.	It is measured by spring balance.
The S.I unit of mass is 'Kg'.	The S.I unit of weight is 'Newton'.
Mass is constant at everywhere.	Weight is variable at different places because it depends on g. i.e, $w = m \times g$
Mass can't be zero.	Weight can be zero.

• **The weight of an object on the moon**

If the mass of an object on the Earth = m kg

& The weight of an object on the Earth = W N

If the weight of an object on the Moon = W' N

If the mass of Earth = M kg

The mass of the moon = M' kg

If the radius of the earth = R

And the radius of the moon = R'

$$\therefore F = G \times \frac{Mm}{R^2}$$

$$\Rightarrow W = G \times \frac{Mm}{R^2} \quad \dots\dots\dots (i) \quad [F = W]$$

Here, Force acts on the object is weight "W". (at earth)

Similarly, weight on the moon

$$W' = G \times \frac{M' M}{(R')^2} \quad \dots\dots\dots (ii)$$

On dividing equation (ii) by equation (i)

$$\Rightarrow \frac{W'}{W} = \frac{\frac{G M' M}{(R')^2}}{\frac{G M m}{R^2}} = \frac{R \times M'}{(R')^2 \times M}$$

$$\Rightarrow W' = W \left\{ \frac{R^2 \times M'}{(R')^2 \times M} \right\}$$

$$\Rightarrow W' = W \left\{ \frac{6400 \times 6400 \times 7.4 \times 10^{22}}{1740 \times 1740 \times 6 \times 10^{24}} \right\}$$

$$\Rightarrow W' = W \times \frac{1}{6}$$

$$\Rightarrow W' = \frac{1}{6} \times W$$

For example:-

Q. If a person of weight 60 N on the earth surface. Find the weight of the same person on the moon.

Sol. We know that the weight of the object on the moon is $\frac{1}{6}$ on the earth.

$$\therefore \text{Weight on the moon} = \frac{1}{6} \times \text{weight on earth}$$

$$= \frac{1}{6} \times 60$$

$$\therefore \text{Weight on the moon} = 10 \text{ N}$$

- **The acceleration due to gravity (g) on the earth**

$$\therefore \text{Weight on the moon} = \frac{1}{6} \times \text{weight on the earth}$$

$$\Rightarrow W' = \frac{1}{6} \times W$$

$$\Rightarrow mg' = \frac{1}{6} \times mg$$

$$\Rightarrow g' = \frac{1}{6} \times g$$

Where, g' is acceleration due to gravity on the moon.

$$\text{i.e, } g' = \frac{1}{6} \times 9.8$$

$$\Rightarrow g' = 1.63 \text{ m/s}^2 \quad [\text{Approx}]$$

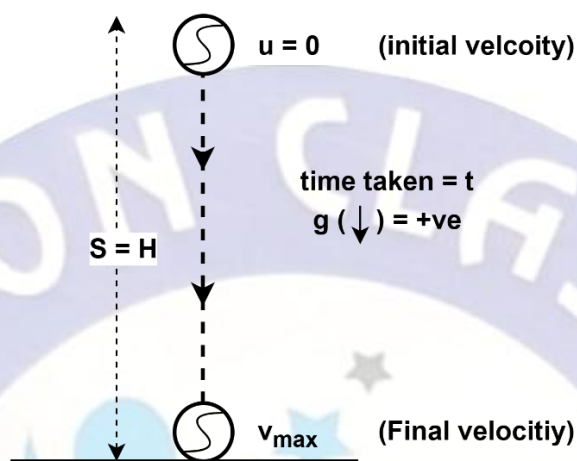
→ Which is required value g on the moon.

For example:-

Q. If a person jumps 2m on the earth then what height jumps on the moon.

Sol. the jumps of the moon $= 2 \times 6$
 $= 12 \text{ m}$

• The equation for free fall objects



1. Time of Descent (नीचे आने में लगा समय)

$$\because v = u + at$$

$$\Rightarrow v = 0 + gt \quad [\because a = g]$$

$$\Rightarrow v = gt$$

Final velocity

$$\Rightarrow t = \frac{v}{g}$$

$$\Rightarrow t_d = \frac{v}{g}$$

Time of descent (नीचे आने में लगा समय)

2. $v^2 = u^2 + 2as$

$$\Rightarrow v^2 = 0^2 + 2gh \quad [\because a = g, S = H]$$

$$\Rightarrow v^2 = 2gh$$

$$\Rightarrow H = \frac{v^2}{2g}$$

Maximum height

3. $S = ut + \frac{1}{2}at^2$

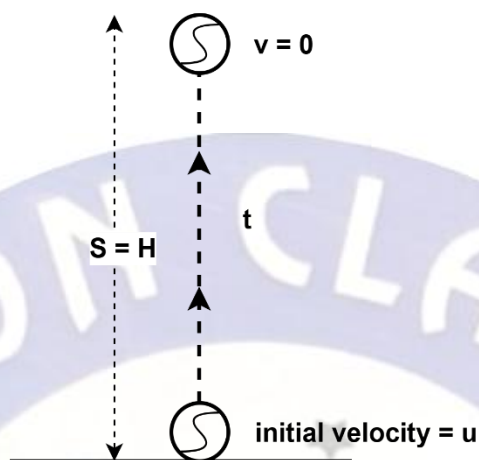
$$\Rightarrow H = 0 \times t + \frac{1}{2}gt^2 \quad [\because H = S, u = 0]$$

$$\Rightarrow H = \frac{1}{2}gt^2$$

• The equation of vertically upward throwing objects

Initial velocity = u

Final velocity (v) = 0



1. Time of Ascent (ऊपर जाने में लगा समय)

$$\because v = u + at$$

$$\Rightarrow 0 = u + (-g) \times t$$

$$\Rightarrow 0 = u - gt$$

$$\Rightarrow gt = u$$

$$[\because a = -g (\uparrow)]$$

$$\Rightarrow t_a = \frac{u}{g}$$

→ Time of Ascent

2. $\because v^2 = u^2 + 2as$

$$\Rightarrow 0^2 = u^2 + 2 \times (-g) \times H \quad [\because a = -g \uparrow \text{ \& } S = H]$$

$$\Rightarrow 0 = u^2 - 2gh$$

$$\Rightarrow 2gh = u^2$$

$$\Rightarrow H = \frac{u^2}{2g}$$

→ Maximum height

3. Time to flying (उड़ान काल)

➤ It is time to fly object in the air = T (Let)

i.e, T = time of ascent + time of descent

$$T = t_a + t_d$$

$$T = \frac{u}{g} + \frac{u}{g}$$

$$\Rightarrow T = \frac{2u}{g}$$

→ Time of flying

Note:

➤ Time of Ascent = Time of descent.

i.e, $t_a = t_d$

- **Pressure**

- The force acted per unit area is called pressure.

i.e, $\text{Pressure} = \frac{\text{Force}}{\text{Area}}$

Pressure is denoted by 'P'.

$$\therefore P = \frac{F}{A}$$

Note:

1. $P \propto F$
2. $P \propto \frac{1}{A}$

The S.I unit of pressure $= \frac{\text{The S.I unit of Force}}{\text{The S.I unit of Area}}$
 $= \frac{\text{Newton}}{(\text{meter})^2} = \text{N/m}^2$

Note:

$$1 \text{ N/m}^2 = 1 \text{ Pascal}$$

$\therefore$ The S.I unit of pressure is 'pascal' and it is denoted by 'Pa'.

Note:

- 'Pascal' Named in honoured of Blaise Pascal (A French mathematician, physicist, philosopher etc).

Note:

1. Effect of any force depends on the area where it is acted.
2. Pressure can be increased by reducing in the contact area.

For example:-

- ✓ Knife, axe, Scissors, Niddle, pin etc may be easily embeded due to pointless and sharpness.

- **Thrust**

- Thrust is a type of force which is acted perpendicularly to the surface.
- The S.I unit of thrust is 'Newton'.

Note:

- The force acted perpendicularly per unit area is called pressure.
i.e, $\text{Pressure} = \frac{\text{Thrust}}{\text{Area}}$

For example:-

Q. If 100 N Force applied on a nail of area 0.0001 m². Find the pressure on the nail.

Ans $\therefore F = 100 \text{ N}, A = 0.0001 \text{ m}^2$

$$\therefore P = \frac{F}{A} = \frac{100}{0.0001} = \frac{100000}{1} = 10^6 \text{ pa}$$

- **Pressure in fluids**

- **Pascal's law**

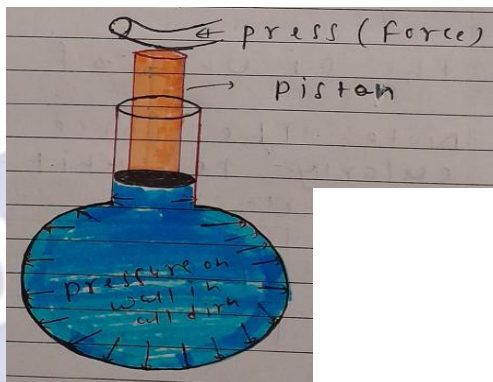
- According to the Pascal's law, "The pressure exerted by the fluids (liquid / gas) in closed container is uniform in all direction as well as wall of the container."

OR

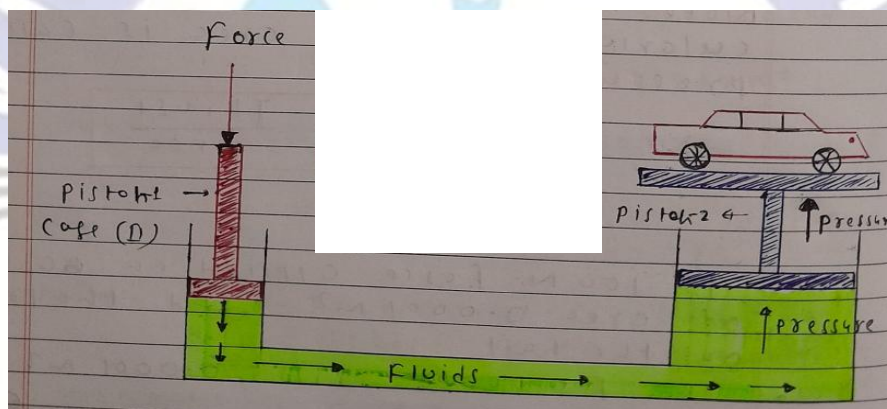
“The pressure applied to an enclosed fluid will be transmitted without a change in magnitude to every point of the fluid and to the walls of the container”

Only for understand

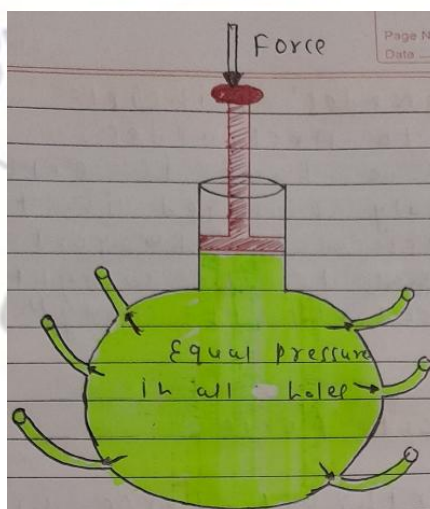
Case 1:



Case 2:



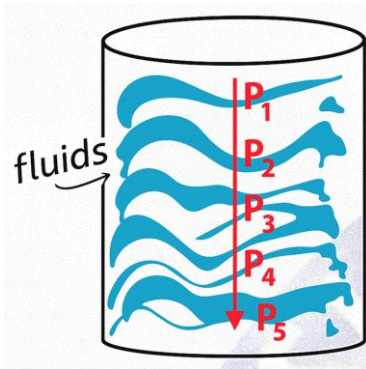
Case 3:



- Application of Pascal's law
 1. In Hydraulic crane.
 2. In Hydraulic lift.
 3. In Hydraulic jack.
 4. In Hydraulic break System etc.

Note:

- Pressure increase in the liquid by increasing the depth.
 i.e, Pressure $\propto$ depth of liquid



Here, $P_1 < P_2 < P_3 < P_4 < P_5$ (due to depth)

When we go depth in the water we feel pressure in our ear drum. Is cause of ear pain

- **Archimedes' Principle**

- According to Archimedes, "When a body is either partially or wholly emerged in the fluid. It experiences a buoyant force which is equal to the weight of liquid displaced by the body".

- **Buoyant Force**

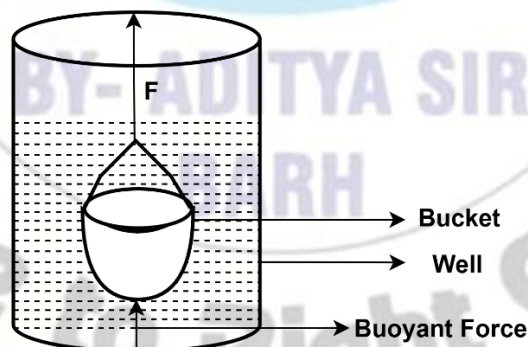
- An upthrust force which is acting on a body emerged in a liquid is called buoyant force.
- The upward force exerted by a liquid is also known as upthrust thus, the buoyant force is an upthrust.

- **According to Archimedes' principle**

- The lackness of weight of body = the weight of liquid displaced by the body

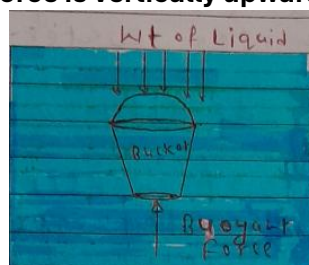
For example:-

- ✓ When a heavy body seems lighter in the water due to a buoyant force.



Note:

- The direction of buoyant Force is vertically upward.



Here, Bucket is in balanced condition because
 weight of water on bucket = buoyant force on the bucket

- **Density**

- The amount of substance per unit volume is called density of that substance.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Note:

1. Density $\propto$ Mass.

2. Density $\propto \frac{1}{\text{Volume}}$

- The S.I unit of density = $\frac{\text{The S.I unit of mass}}{\text{The S.I unit of volume}} = \frac{\text{Kg}}{\text{m}^3}$
 So, The S.I unit of density = kg /m³

- The density of sea water is more than density of normal water due to which swimming is more easier in sea water.

- **Relative density**

- The ratio of density of any substance and density of water at 4°C is known as relative density.

i.e,

$$\text{Relative density} = \frac{\text{Density of any substance}}{\text{Density of water at } 4^{\circ}\text{C}}$$

- The relative density is ratio of densities so, it is a unitless quantity.
- Density and relative density is measured by “Hydrometer”.

Note:

- Density of any substance = Relative density $\times$ density of water at 4°C

- **Differences between density and relative densities**

1. Density has unit kg/m³ but relative density is a unitless quantity.
2. Density = $\frac{\text{mass}}{\text{volume}}$ while relative density is a ratio of densities.

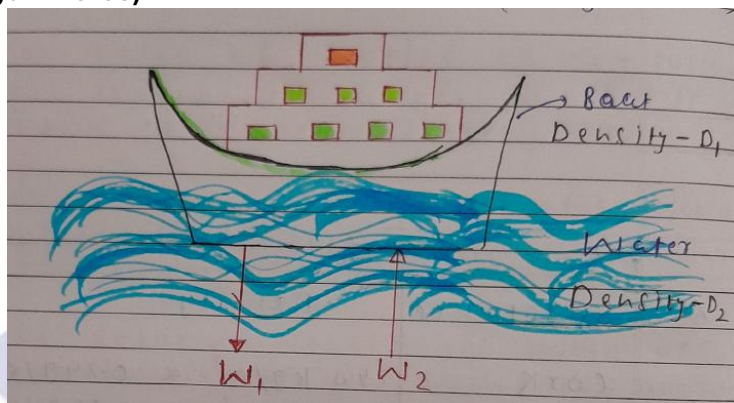
Substance	Density	Relative Density
Cork	240 kg/m ³ OR 0.24 g/cm ³	0.24
Wood	600 kg/m ³ OR 0.88 g/cm ³	0.8
Ice	920 kg/m ³ OR 0.92 g/cm ³	0.92
Water	1000 kg/m ³ OR 1.0 g/cm ³	1.0
Glass	2500 kg/m ³ OR 2.5 g/cm ³	2.5
Aluminum	2700 kg/m ³ OR 2.7 g/cm ³	2.7
Iron	7800 kg/m ³ OR 7.8 g/cm ³	7.8
Mercury	13600 kg/m ³ OR 13.6 g/cm ³	13.6
Gold	19300 kg/m ³ OR 19.3 g/cm ³	19.3
Glycerin	1260 kg/m ³ OR 1.26 g/cm ³	1.26

Note:

- To change kg/m³ into g/cm³ then multiply with $\frac{1}{1000}$.

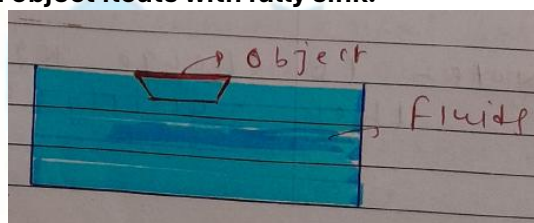
- **Floating of objects**

- Floating of an object experience two forces,
 1. Towards downward weight (W_1).
 2. Upward force W_2 (Buoyant Force).



Case 1: If $W_1 > W_2$ i.e, $D_1 > D_2$ then boat (any object) sinks completely.

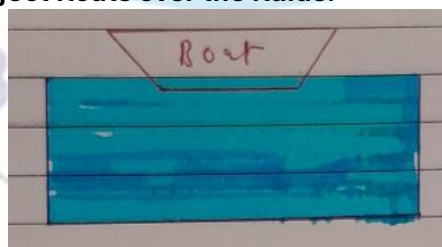
Case 2: If $W_1 = W_2$ i.e, $D_1 = D_2$ then object floats with fully sink.



Note:

- In this case one part of Ice (object) above the water and ninth parts below the water. $\left(\frac{1}{2}\right)$

Case 3: If $W_1 < W_2$ i.e, $D_1 < D_2$ then object floats over the fluids.



Note:

- $1 \text{ kg wt} = 1 \text{ kg} \times 9.8 \text{ m/s}^2 = 9.8 \text{ N}$
 $\therefore 1 \text{ kg wt} = 9.8 \text{ N}$
- The mass of 1 kg object is known as 1 kg wt object, and the force is called gravitational unit.
i.e, $1 \text{ kg wt} = 1 \text{ kg} \times 9.8 \text{ m/s}^2 = 9.8 \text{ N}$

Q. An iron's nails sinks in the water but an iron boat floats over water why ?

Ans because the density of iron nails is greater than the water so, it sinks while the density of boat is less than the water so it floats.